

Observable-Class Derivation Mismatch and the Exact Growth Equation: From Proxy Capacity to Block-Level Dynamics

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Abstract

The O3–O7 derivation chain establishes the structural relation $\beta^* = 1/(\delta + 1/2)$ linking the capacity decay exponent δ to the cascade growth exponent β^* governing charged-lepton mass ratios. Papers O12–O14 demonstrated that the exact Weil-block capacity exponent $\hat{\delta}_{\text{exact}}(q) \in [3.6, 4.8]$ is incompatible with the target window $\delta \in [7.4, 10.6]$, and that this tension is structural rather than a finite-size artefact. The present paper identifies the precise origin of the mismatch: the O6 growth equation $dp/dn \lesssim R_n c_\chi \sqrt{p(n)}$ derives β^* from a *scalar global* redundancy functional R_n , whereas the exact observable measured in O12–O13 is a *block-averaged* capacity over $(q - 1)$ Weil representations of internal dimension q . We prove a non-equivalence theorem establishing that the block-averaged exponent $\hat{\delta}_{\text{exact}}$ does not, in general, coincide with the scalar exponent α entering the growth equation. We define the effective dynamical redundancy R_n^{eff} as the observable that genuinely governs the valence growth in the exact-block setting, and re-derive the growth equation with this observable as input. The re-derivation yields two structural scenarios: (C1) the functional form $\beta^* = 1/(\delta_{\text{eff}} + 1/2)$ survives with a corrected exponent δ_{eff} that differs from $\hat{\delta}_{\text{exact}}$ by an aggregation bias; (C2) the functional form itself is modified by a representation-dimension correction, $\beta^* = 1/(\delta_{\text{eff}} + 1/2 + \sigma(q))$, where $\sigma(q) \rightarrow 0$ as $q \rightarrow \infty$. Scenario discrimination requires a block-level redundancy measurement not yet available from the O12 pipeline, identified as the central open direction of this paper.

Keywords. Observable-class mismatch, growth equation, block-averaged capacity, scalar redundancy, Weil representations, Heisenberg graphs, cascade exponent, spectral admissibility.

Contents

1	Introduction	4
1.1	Position in the O-series	4
1.2	The central question	4
1.3	Main results	4
2	Audit of the O3–O7 Derivation Chain	4
2.1	Step A: O3 mass formula	5
2.2	Step B: O6 growth equation	5
2.3	Step C: O7 reinterpretation	5
2.4	Summary of audit	5
3	Non-Equivalence of Block-Averaged and Scalar Exponents	6
3.1	Setup	6
3.2	Block heterogeneity and exponent bias	6
3.3	Consequence for the growth equation	7
4	Effective Dynamical Redundancy	7
4.1	Physical meaning of the growth equation	7
4.2	Definition of R_n^{eff}	7
4.3	Exponent of R_n^{eff}	8
5	Re-Derivation of the Growth Equation	9
5.1	Growth equation in the exact-block setting	9
5.2	Separation of variables	9
5.3	Two structural scenarios	10
6	Scenario Discrimination	10
6.1	What would discriminate C1 from C2	10
6.2	Required measurement: block-level redundancy	11
7	Asymptotic Analysis	11
7.1	The $q \rightarrow \infty$ limit	11
7.2	Finite- q phenomenology	11
8	Open Directions	12
9	Conclusion	12

Notation glossary

The four exponents appearing in this paper form a strict hierarchy:

Symbol	Name	Definition
$\hat{\delta}_{\text{exact}}$	measured exponent	Power-law fit of the unweighted block average $\bar{\Sigma}_n = \mathbb{E}_c[\Sigma_n^{(c)}] \sim A n^{-\hat{\delta}_{\text{exact}}}$ (O12/O13 observable).
α_{dyn}	dynamical exponent	Power-law fit of the weighted observable $R_n^{\text{eff}} \sim B n^{-\alpha_{\text{dyn}}}$ (Definition 4.4). Satisfies $\alpha_{\text{dyn}} \leq \hat{\delta}_{\text{exact}}$ (Corollary 4.5).
δ_{eff}	effective exponent (C1)	Redefined exponent absorbing shell-growth and q -normalisation corrections: $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - (D - 1) + \log q / \log n^*$.
$\sigma(q)$	structural correction (C2)	Additional q -dependent term in the denominator of the growth formula, required if C1 does not close the gap to $\beta^* \in (0.09, 0.13)$.

1 Introduction

1.1 Position in the O-series

The admissibility sub-programme has traced a complete arc from obstruction identification (O8–O9 [13, 14]) through exact measurement (O10–O13 [15, 16, 17, 18]) to theoretical correspondence (O14 [19]). O14 established that the tension between the exact Weil-block capacity exponent $\hat{\delta}_{\text{exact}}(q) \in [3.6, 4.8]$ and the phenomenological target $\delta \in [7.4, 10.6]$ is not a numerical artefact (refuting hypothesis S1 of O13) but a structural mismatch (Source S2). The corrected relation derived in O14,

$$\beta^* = \frac{1}{\delta_{\text{eff}} + \frac{1}{2}} + \epsilon(q), \quad (1)$$

with $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - \delta_{\gamma, \text{corr}} - \eta \cdot \log q / \log n^*$ and $\epsilon(q) \rightarrow 0$ as $q \rightarrow \infty$, was shown to be consistent with Scenario S2-B for all tested primes: $\delta_{\text{eff}} < 5.0$ throughout, with the normalisation correction $\eta \cdot \log q / \log n_1 \approx 1.03\text{--}1.06$ as the dominant term.

O14 thus closed the measurement question but left open the *derivation* question: does the functional form $\beta^* = 1/(\delta + 1/2)$ itself remain valid when the input observable changes from a scalar global capacity to a block-averaged Weil capacity? The present paper answers this question.

1.2 The central question

The relation $\beta^* = 1/(\delta + 1/2)$ was derived in O6 [11] (Proposition 4.2) from the growth equation

$$\frac{dp}{dn} \lesssim R_n \cdot c_\chi \sqrt{p(n)}, \quad (2)$$

where R_n is a scalar redundancy functional satisfying $R_n \sim p(n)^{-\alpha}$ in the pre-saturation window. Separation of variables yields $p(n) \sim n^{1/(\alpha+1/2)}$, giving $\beta^* = 1/(\alpha + 1/2)$.

The question is: *what is the correct identification of R_n in the exact Weil-block setting, and does substituting it into (2) preserve the functional form?*

1.3 Main results

- (1) We audit the O3–O7 derivation chain and identify the three precise points where the scalar-observable hypothesis enters (Section 2).
- (2) We prove a non-equivalence theorem: the block-averaged exponent $\hat{\delta}_{\text{exact}}$ does not coincide with the scalar exponent α of the growth equation under generic conditions on block heterogeneity (Section 3).
- (3) We define the effective dynamical redundancy R_n^{eff} and show it is the correct input to (2) in the exact setting (Section 4).
- (4) We re-derive the growth equation with R_n^{eff} as input and identify the two structural scenarios C1 and C2 (Section 5).
- (5) We specify the block-level measurement required to discriminate C1 from C2 (Section 6).

2 Audit of the O3–O7 Derivation Chain

We trace the logical chain that produces $\beta^* = 1/(\delta + 1/2)$ and identify, at each step, the hypotheses that hold for the proxy observable but fail for the exact Weil-block observable.

2.1 Step A: O3 mass formula

O3 [8] derives the inter-generation mass ratio

$$\frac{M_i}{M_j} = \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|} \right)^{2/\beta}, \quad (3)$$

from the support-narrowing mechanism under the power-law growth hypothesis $p(n) \sim An^\beta$. The derivation uses two hypotheses:

H-O3-1

The growth $p(n) \sim An^\beta$ is a power law.

H-O3-2

The mass is inversely proportional to the exit rank: $M_i \propto E_P/n_{\text{exit}}(\lambda_i)$.

Both hypotheses are explicit structural postulates of O3. Neither involves any property of the capacity observable. The mass formula (3) is therefore *independent* of the observable-class mismatch and remains valid throughout.

2.2 Step B: O6 growth equation

O6 [11], Proposition 4.2, derives $\beta^* = 1/(\alpha + 1/2)$ from the growth equation (2). The derivation involves three implicit hypotheses about the redundancy functional R_n :

H-O6-1 (Scalar observable)

R_n is a single real-valued function of the cascade rank n , independent of any internal representation label.

H-O6-2 (Power-law decay)

$R_n \sim p(n)^{-\alpha}$ for a single exponent α over the pre-saturation window.

H-O6-3 (Direct coupling)

R_n enters the growth equation multiplicatively: the rate of valence increase is proportional to R_n .

Hypothesis H-O6-1 is the central point of failure. In the proxy setting (O7–O11), the capacity C_n^{proxy} is defined via a Fourier-character proxy that abelianises the representation structure [12]. It is indeed a scalar function of n .

In the exact setting (O12–O14), the observable is $\bar{\Sigma}_n = \mathbb{E}_c[\Sigma_n^{(c)}]$, a mean over $(q-1)$ Weil blocks, each of internal dimension q . This is not a scalar observable in the sense of H-O6-1: it carries a hidden dependence on the block structure through the averaging.

2.3 Step C: O7 reinterpretation

O7 [12] recasts the O6 formula in capacity language. The identification $\alpha = \delta$ (O7, Section 5.1) holds in the reduced filling model where $p(n) = n$. The state law $R_n^{(k)} \approx \Phi(\eta_n)$ with $\Phi(\eta) = 1/\sqrt{1+\eta^2}$ is derived exactly in the reduced model and is a property of the scalar capacity.

O7 does not introduce new deductive content for the $\delta \rightarrow \beta^*$ relation. It inherits H-O6-1 from O6 without modification. The Remark 5.2 of O7 states this explicitly: the formula “does not introduce a new derivation of β : it recasts the O6 target in capacity language without circularity.”

2.4 Summary of audit

Table 1: Audit of the derivation chain. The column “Exact status” indicates whether the hypothesis holds for the exact Weil-block observable.

Paper	Hypothesis	Proxy	Exact
O3	H-O3-1 (power-law growth)	valid	valid
O3	H-O3-2 (mass $\propto 1/n_{\text{exit}}$)	valid	valid
O6	H-O6-1 (scalar observable)	valid	fails
O6	H-O6-2 (power-law decay)	valid	valid
O6	H-O6-3 (direct coupling)	valid	open
O7	inherits H-O6-1	valid	fails

The mass formula of O3 is intact. The growth equation of O6 is algebraically correct in its domain. The failure is localised at H-O6-1: the scalar-observable hypothesis, which enters the growth equation through the identification of R_n with a single power-law observable.

3 Non-Equivalence of Block-Averaged and Scalar Exponents

We now prove that the exponent extracted from a block-averaged observable does not, in general, coincide with the exponent of a scalar observable satisfying the growth equation.

3.1 Setup

Fix an odd prime q . The Heisenberg group $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ has $(q-1)$ faithful irreducible representations ρ_c ($c \in (\mathbb{Z}/q\mathbb{Z})^\times$), each of dimension q . The block-level projective capacity at BFS depth n in block c is

$$\Sigma_n^{(c)} = \frac{\Delta r_n^{(c)}}{|S_n|}, \quad (4)$$

where $\Delta r_n^{(c)}$ is the incremental projective span in representation ρ_c and $|S_n|$ is the BFS shell size. The block average is $\bar{\Sigma}_n = \mathbb{E}_c[\Sigma_n^{(c)}]$.

In the proxy setting, the capacity is a single scalar C_n^{proxy} defined via the Fourier-character abelianisation [16].

3.2 Block heterogeneity and exponent bias

Definition 3.1 (Block exponent and mean exponent). *For each block c , define the block exponent $\delta^{(c)}$ by the power-law fit $\Sigma_n^{(c)} \sim A_c n^{-\delta^{(c)}}$ over the fitting window $[n_0, n_1]$. Define the mean exponent $\bar{\delta} = \mathbb{E}_c[\delta^{(c)}]$ and the exponent of the mean $\hat{\delta}_{\text{exact}}$ by the fit $\bar{\Sigma}_n \sim A n^{-\hat{\delta}_{\text{exact}}}$.*

Proposition 3.2 (Non-equivalence). *If the block amplitudes A_c are not all equal, then $\hat{\delta}_{\text{exact}} \neq \bar{\delta}$ in general. Specifically, let $V_A = \text{Var}_c(A_c)/\mathbb{E}_c[A_c]^2$ be the relative amplitude variance. Then*

$$\hat{\delta}_{\text{exact}} = \bar{\delta} - \frac{1}{\log(n_1/n_0)} \log \left(\frac{\mathbb{E}_c[A_c (n_1/n_0)^{-\delta^{(c)}}]}{\mathbb{E}_c[A_c] \cdot (n_1/n_0)^{-\bar{\delta}}} \right). \quad (5)$$

The correction term vanishes if and only if $\delta^{(c)}$ is independent of c (block universality).

Proof. Write $\bar{\Sigma}_n = \mathbb{E}_c[A_c n^{-\delta^{(c)}}]$. At two endpoints n_0, n_1 of the fitting window:

$$\hat{\delta}_{\text{exact}} = -\frac{\log(\bar{\Sigma}_n[n_1]/\bar{\Sigma}_n[n_0])}{\log(n_1/n_0)} = -\frac{1}{\log(n_1/n_0)} \log \left(\frac{\mathbb{E}_c[A_c n_1^{-\delta^{(c)}}]}{\mathbb{E}_c[A_c n_0^{-\delta^{(c)}}]} \right).$$

If all $\delta^{(c)} = \delta_0$, this reduces to $\delta_0 = \bar{\delta}$. Otherwise, the ratio is a weighted average of $\exp(-\delta^{(c)} \log(n_1/n_0))$ with weights A_c , and by Jensen's inequality it differs from $\exp(-\bar{\delta} \log(n_1/n_0))$ whenever the distribution of $\delta^{(c)}$ is non-degenerate. $\square$

Remark 3.3 (Direction of the bias). *For a convex function $f(\delta) = \exp(-\delta \log(n_1/n_0))$, Jensen's inequality gives $\mathbb{E}_c[f(\delta^{(c)})] \geq f(\bar{\delta})$, so $\hat{\delta}_{\text{exact}} \leq \bar{\delta}$. The block-averaged exponent is systematically smaller than the mean of individual block exponents. This is consistent with the O12–O13 observation that $\hat{\delta}_{\text{exact}}$ lies below the target window.*

Remark 3.4 (Quantitative estimate from O12–O13 data). *O12 [17] reports inter-block variance ratios $V_n^{\text{max}}(q)$ that decrease from ~ 0.15 at $q = 29$ to ~ 0.03 at $q = 211$. Even at $q = 211$, the residual variance produces a bias of order $|\hat{\delta}_{\text{exact}} - \bar{\delta}| \sim 0.1\text{--}0.3$, which is insufficient to bridge the gap to the target $\delta \in [7.4, 10.6]$. The non-equivalence is therefore real but quantitatively small compared to the total deficit.*

3.3 Consequence for the growth equation

The growth equation (2) couples the rate of valence increase to the redundancy of new relational directions. Physically, it is the *fraction of genuinely new directions contributed by the BFS frontier* that limits the growth of p . This fraction is a property of the *aggregate* projective dynamics, not of any individual block.

The scalar exponent α in H-O6-2 is the exponent of this aggregate fraction. It is not, in general, equal to $\hat{\delta}_{\text{exact}}$ (the exponent of the block-averaged capacity), because the block-averaged capacity includes internal normalisation and phase structure that do not enter the growth equation.

This motivates the definition of the effective dynamical redundancy in the next section.

4 Effective Dynamical Redundancy

4.1 Physical meaning of the growth equation

The growth equation (2) has a precise physical content: at each cascade step n , the substrate attempts to increase its effective relational valence p by incorporating new relational directions from the BFS frontier. The rate of successful incorporation is limited by two factors:

- (a) The Born–Infeld flux bound c_χ , which caps the rate of change of any relational variable [9].
- (b) The fraction of frontier directions that are genuinely new, i.e. not already represented in the current relational structure.

Factor (b) is what R_n encodes. In the proxy setting, where the capacity is a scalar, R_n is directly proportional to the capacity: $R_n \propto \Sigma_n$. In the exact setting, the identification is more subtle because “genuinely new” must be defined relative to the full block structure.

4.2 Definition of R_n^{eff}

Definition 4.1 (Effective dynamical redundancy). *The effective dynamical redundancy at cascade rank n is*

$$R_n^{\text{eff}} = \frac{1}{q-1} \sum_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} w_c(n) \Sigma_n^{(c)}, \quad (6)$$

where $w_c(n)$ is the dynamical weight of block c , defined as

$$w_c(n) = \frac{\Delta r_n^{(c)} / \dim V_c}{\sum_{c'} \Delta r_n^{(c')} / \dim V_{c'}} = \frac{\Delta r_n^{(c)} / q}{\sum_{c'} \Delta r_n^{(c')} / q} = \frac{\Delta r_n^{(c)}}{\sum_{c'} \Delta r_n^{(c')}}. \quad (7)$$

The weight $w_c(n)$ measures the relative contribution of block c to the total incremental span at step n . It is the natural dynamical weight because the growth equation counts the total number of new relational degrees of freedom, and each block contributes $\Delta r_n^{(c)}$ such degrees.

Proposition 4.2 (Relation to the unweighted mean). *Let $\bar{\Sigma}_n = \mathbb{E}_c[\Sigma_n^{(c)}]$ be the unweighted block average (the O12/O13 observable) and let $\Sigma_n^w = \sum_c w_c \Sigma_n^{(c)}$ be the weighted observable of Definition 4.1. Then*

$$R_n^{\text{eff}} = \bar{\Sigma}_n + \text{Cov}_c \left(\frac{w_c}{1/(q-1)}, \Sigma_n^{(c)} \right), \quad (8)$$

where the covariance is taken over the uniform distribution on blocks. The correction vanishes if the weights are uniform ($w_c = 1/(q-1)$ for all c), which occurs if and only if all blocks contribute equally to the incremental span.

Proof. Direct computation: $R_n^{\text{eff}} - \bar{\Sigma}_n = \sum_c w_c \Sigma_n^{(c)} - \frac{1}{q-1} \sum_c \Sigma_n^{(c)} = \sum_c (w_c - \frac{1}{q-1}) \Sigma_n^{(c)} = (q-1) \text{Cov}_c(w_c, \Sigma_n^{(c)})$. $\square$

Remark 4.3 (Sign of the correction). *Blocks with larger $\Delta r_n^{(c)}$ (hence larger w_c) tend to have larger $\Sigma_n^{(c)}$ (since $\Sigma_n^{(c)}$ is the ratio $\Delta r_n^{(c)} / |S_n|$). The covariance is therefore generically positive, giving $R_n^{\text{eff}} \geq \bar{\Sigma}_n$. The effective dynamical redundancy is larger than the unweighted block average.*

4.3 Exponent of R_n^{eff}

Definition 4.4 (Dynamical exponent). *The dynamical exponent α_{dyn} is defined by the power-law fit $R_n^{\text{eff}} \sim B n^{-\alpha_{\text{dyn}}}$ over the pre-saturation window.*

By Proposition 4.2 and Remark 4.3, $\alpha_{\text{dyn}} \leq \hat{\delta}_{\text{exact}}$: the dynamical exponent is smaller than or equal to the unweighted block-average exponent.

Corollary 4.5 (Aggregation no-go). *No choice of dynamical weights $w_c(n)$ satisfying $w_c \geq 0$ and $\sum_c w_c = 1$ can produce $\alpha_{\text{dyn}} > \hat{\delta}_{\text{exact}}$. In particular, the block-averaged exponent $\hat{\delta}_{\text{exact}} \in [3.6, 4.8]$ measured in O12–O13 is an upper bound on the dynamical exponent entering the growth equation. Since $\hat{\delta}_{\text{exact}} < 5.0 \ll 7.4$, no aggregation or reweighting of the existing block-level data can bring the dynamical exponent into the target window $[7.4, 10.6]$.*

Proof. At fixed BFS depth n , the shell size $|S_n|$ is common to all blocks. Therefore $\Sigma_n^{(c)} = \Delta r_n^{(c)} / |S_n|$ is a monotone increasing function of $\Delta r_n^{(c)}$ alone. The weight $w_c = \Delta r_n^{(c)} / \sum_{c'} \Delta r_n^{(c')}$ is likewise a monotone increasing function of the same variable $\Delta r_n^{(c)}$. Since both w_c and $\Sigma_n^{(c)}$ are monotone increasing functions of a single common variable, the Chebyshev sum inequality gives

$$\sum_c w_c \Sigma_n^{(c)} \geq \left(\sum_c w_c \right) \left(\frac{1}{q-1} \sum_c \Sigma_n^{(c)} \right) = \bar{\Sigma}_n,$$

so $R_n^{\text{eff}} \geq \bar{\Sigma}_n$ pointwise in n . A pointwise lower bound on the observable translates into an upper bound on the decay exponent: $\alpha_{\text{dyn}} \leq \hat{\delta}_{\text{exact}}$. $\square$

This is the sharpest negative result of the paper: it eliminates the entire class of solutions based on “better averaging of the blocks.” If the δ - β^* tension is to be resolved, the resolution must come from a modification of the growth equation itself — through the shell growth factor, the q -normalisation, or a structural correction $\sigma(q)$ — not from a refined treatment of the block statistics.

5 Re-Derivation of the Growth Equation

5.1 Growth equation in the exact-block setting

In the exact setting, the BFS frontier at step n contributes new relational degrees of freedom through all $(q-1)$ Weil blocks simultaneously. The total incremental span is

$$\Delta r_n^{\text{tot}} = \sum_c \Delta r_n^{(c)}. \quad (9)$$

Each block c lives in a representation space of dimension q , so the total available representation space has dimension $(q-1) \cdot q \sim q^2$.

The Born–Infeld growth bound becomes

$$\frac{dp}{dn} \lesssim c_\chi \sqrt{p(n)} \cdot \frac{\Delta r_n^{\text{tot}}}{(q-1) \cdot q}, \quad (10)$$

where the denominator $(q-1) \cdot q$ normalises by the total number of representation degrees of freedom.

Remark 5.1 (Status of the q -dependent normalisation). *The factor $1/q$ in (10) is a dimensional ansatz, not a derived result. Its motivation is as follows: in the proxy setting, the capacity is defined in a scalar (one-dimensional) representation, so no internal dimension enters the normalisation. In the exact setting, each block has q internal degrees of freedom, and the fraction of genuinely new directions per available degree of freedom naturally acquires a factor $1/q$. This is consistent with the normalisation correction identified in O14 as the dominant source of the observable-class mismatch. However, the Born–Infeld constraint $|\partial_t \chi_v| \leq c_\chi$ operates at the level of individual relational variables, not at the representation level. The correct translation of this per-variable bound into a per-representation constraint depends on how the representation degrees of freedom couple to the relational dynamics, which remains an open derivation. The factor $1/q$ is the simplest dimensional estimate; a more refined treatment may yield a different power $q^{-\eta}$ with $\eta \neq 1$.*

5.2 Separation of variables

Rewrite (10) using $\Delta r_n^{\text{tot}}/((q-1) \cdot q) = \frac{1}{q} \cdot \frac{1}{q-1} \sum_c \Delta r_n^{(c)}/|S_n| \cdot |S_n| = \frac{|S_n|}{q} \cdot \bar{\Sigma}_n$, where we used $\bar{\Sigma}_n = \frac{1}{q-1} \sum_c \Sigma_n^{(c)}$ with $\Sigma_n^{(c)} = \Delta r_n^{(c)}/|S_n|$. Then

$$\frac{dp}{dn} \lesssim c_\chi \sqrt{p(n)} \cdot \frac{|S_n|}{q} \cdot \bar{\Sigma}_n. \quad (11)$$

In the BFS growth regime on the Heisenberg graph, $|S_n| \sim C_S n^{D-1}$ with D the effective growth dimension, and $\bar{\Sigma}_n \sim A n^{-\hat{\delta}_{\text{exact}}}$. The factor $|S_n| \cdot \bar{\Sigma}_n \sim C n^{D-1-\hat{\delta}_{\text{exact}}}$.

For the proxy derivation (O6), $|S_n|$ is absorbed into the definition of R_n and does not appear explicitly. The exact derivation makes the shell growth and the capacity decay visible as separate factors.

Hypothesis 5.2 (Effective growth law under exact-block normalisation). *Under the ansatz $p(n) \sim A n^\beta$, the exact-block growth equation (11), and the dimensional normalisation ansatz $1/q$ of Remark 5.1, separation of variables gives*

$$\beta = \frac{1}{\frac{1}{2} + \hat{\delta}_{\text{exact}} - (D-1) + \log q / \log n^*}, \quad (12)$$

where n^* is the saturation depth. In the limit $q \rightarrow \infty$ with $n^* \sim q$, the term $\log q / \log n^* \rightarrow 1$.

Remark 5.3 (Status of Hypothesis 5.2). *This is a heuristic re-derivation, not a theorem. Two inputs remain conjectural: (i) the normalisation factor $1/q$ in (10) is a dimensional ansatz motivated by the representation structure (each block has q internal degrees of freedom), but it has not been derived from the Born–Infeld constraint in the block setting; (ii) the term $\log q / \log n^*$ is an effective translation of the q -dependent normalisation into exponent space, valid only at the characteristic depth n^* . What is rigorously established is that the proxy formula $\beta = 1/(1/2 + \delta)$ cannot be applied to $\hat{\delta}_{\text{exact}}$ without modification; the precise form of the modification is the content of this hypothesis.*

Derivation sketch. Substituting $p(n) = A n^\beta$ and the power-law forms into (11):

$$A \beta n^{\beta-1} \lesssim c_\chi \sqrt{A} n^{\beta/2} \cdot \frac{C}{q} n^{D-1-\hat{\delta}_{\text{exact}}}.$$

Collecting powers of n : $\beta - 1 = \beta/2 + D - 1 - \hat{\delta}_{\text{exact}} - \log q / \log n$, where $\log q / \log n$ accounts for the q -dependent normalisation evaluated at the characteristic depth $n = n^*$. Solving for β : $\beta/2 = \hat{\delta}_{\text{exact}} - D + 1 + \log q / \log n^*$, whence $\beta = 1/(\frac{1}{2} + \hat{\delta}_{\text{exact}} - (D-1) + \log q / \log n^*)$. In the proxy limit ($q = 1$, $D = 1$), this reduces to $\beta = 1/(1/2 + \delta)$, recovering the O6 formula.

5.3 Two structural scenarios

Scenario C1: Same functional form, corrected exponent

If the shell growth exponent $D - 1$ and the normalisation term $\log q / \log n^*$ combine to produce an effective correction that can be absorbed into a redefined exponent $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - (D - 1) + \log q / \log n^*$, then (12) takes the form

$$\beta^* = \frac{1}{\delta_{\text{eff}} + \frac{1}{2}}, \quad (13)$$

with δ_{eff} carrying the full correction. This is the scenario where the O6 formula survives structurally but with a different numerical input.

For the Heisenberg graph with $D \approx 4$ (Bass–Guivarc’h growth exponent), $(D - 1) = 3$, and $\log q / \log n^* \approx 1$ at $n^* \sim q$, one obtains $\delta_{\text{eff}} \approx \hat{\delta}_{\text{exact}} - 3 + 1 = \hat{\delta}_{\text{exact}} - 2$. With $\hat{\delta}_{\text{exact}} \in [3.6, 4.8]$, this gives $\delta_{\text{eff}} \in [1.6, 2.8]$, yielding $\beta^* \in [0.26, 0.48]$. This is still above the target $(0.09, 0.13)$.

Scenario C2: Modified functional form

If the corrections cannot be absorbed into a single redefined exponent, the growth equation acquires an explicit q -dependent structural term:

$$\beta^* = \frac{1}{\delta_{\text{eff}} + \frac{1}{2} + \sigma(q)}, \quad (14)$$

where $\sigma(q)$ encodes the residual effect of the block structure on the growth dynamics and satisfies $\sigma(q) \rightarrow 0$ as $q \rightarrow \infty$.

For C2 to produce $\beta^* \in (0.09, 0.13)$ with $\delta_{\text{eff}} \in [1.6, 2.8]$, the correction must satisfy $\sigma(q) \in [5.1, 8.6]$, which would require $\sigma(q) \sim q^\kappa$ for some $\kappa > 0$. Whether such a correction arises from the Weil-block representation structure is the central open question.

6 Scenario Discrimination

6.1 What would discriminate C1 from C2

The two scenarios make different predictions for the q -dependence of the dynamical exponent α_{dyn} :

Under C1, α_{dyn} absorbs all corrections and its q -dependence is fully captured by the shift $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - 2$. A measurement of R_n^{eff} (Definition 4.1) at multiple primes would show power-law decay with an exponent that *stabilises* as q grows.

Under C2, α_{dyn} stabilises but $\sigma(q)$ introduces a separate q -dependent correction. The growth equation would show a systematic q -dependent departure from the C1 prediction.

6.2 Required measurement: block-level redundancy

The O12/O13 pipeline [17, 18] computes $\Sigma_n^{(c)}$ per block but discards the complex fingerprint vectors $z_b^{(c)}(n)$ after Gram–Schmidt orthogonalisation. The weights $w_c(n)$ of Definition 4.1 are computable from the existing pipeline outputs ($\Delta r_n^{(c)}$ is stored).

The block-level redundancy measurement therefore requires:

- (i) Compute $w_c(n)$ from the stored $\Delta r_n^{(c)}$ values.
- (ii) Form $R_n^{\text{eff}} = \sum_c w_c(n) \Sigma_n^{(c)}$.
- (iii) Fit $R_n^{\text{eff}} \sim B n^{-\alpha_{\text{dyn}}}$ and extract α_{dyn} .
- (iv) Compare α_{dyn} with $\hat{\delta}_{\text{exact}}$ to measure the aggregation bias.
- (v) Test whether $\beta = 1/(\alpha_{\text{dyn}} + 1/2)$ falls in $(0.09, 0.13)$ (C1), or whether an additional q -dependent term is required (C2).

Step (i) is immediately feasible from existing O12/O13 data. Steps (ii)–(v) constitute the minimal computational programme for discriminating the two scenarios.

7 Asymptotic Analysis

7.1 The $q \rightarrow \infty$ limit

In the limit $q \rightarrow \infty$, three simplifications occur:

- (a) Block universality: $\text{Var}_c(\Sigma_n^{(c)}) \rightarrow 0$ by the O13 variance reduction result [18], so $R_n^{\text{eff}} \rightarrow \bar{\Sigma}_n$ and $\alpha_{\text{dyn}} \rightarrow \hat{\delta}_{\text{exact}}$.

(b) The normalisation correction $\log q / \log n^* \rightarrow 1$.

(c) The correction $\epsilon(q) \rightarrow 0$ in (1).

In this limit, the growth equation reduces to $\beta = 1/(1/2 + \hat{\delta}_{\text{exact}} - 2)$ under C1, or $\beta = 1/(1/2 + \hat{\delta}_{\text{exact}} - 2 + \sigma(\infty))$ under C2.

If $\sigma(\infty) = 0$, C2 reduces to C1 asymptotically. If $\sigma(\infty) > 0$, C2 introduces a permanent structural correction even in the continuum limit.

7.2 Finite- q phenomenology

At accessible primes ($q \leq 211$), the correction terms are non-negligible. Table 2 reports the predicted β^* under both scenarios for the five O12/O13 primes.

Table 2: Predicted β^* under scenarios C1 and C2 for the five O12/O13 primes. C1 uses $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - 2$; C2 requires $\sigma(q)$ to reach the target.

q	$\hat{\delta}_{\text{exact}}$	δ_{eff} (C1)	β^* (C1)	$\sigma(q)$ needed (C2)
29	4.37	2.37	0.348	5.4
61	4.80	2.80	0.303	5.9
101	4.28	2.28	0.360	5.3
151	3.90	1.90	0.417	5.0
211	3.59	1.59	0.478	4.7

Under C1, $\beta^* \in [0.30, 0.48]$ for all tested primes: closer to the target than the uncorrected value $\beta^* \in [0.19, 0.21]$ from O12, but still above $(0.09, 0.13)$ by a factor of approximately 3.

Under C2, the required correction $\sigma(q) \in [4.7, 5.9]$ is large and shows a mild q -dependence. Whether a correction of this magnitude can arise from the Weil-block structure is an open question.

8 Open Directions

O15-O1 (Block-level redundancy extraction)

Compute R_n^{eff} from the O12/O13 pipeline outputs and extract α_{dyn} at $q \in \{29, 61, 101, 151, 211\}$. This is the minimal measurement required to discriminate C1 from C2.

O15-O2 (Shell growth exponent)

The effective growth dimension $D \approx 4$ (Bass–Guivarc’h) is an asymptotic estimate. The exact value of D in the pre-saturation window $[n_0, n_1]$ at finite q may differ. Extracting D directly from the BFS data would sharpen the C1 prediction.

O15-O3 (Representation-theoretic $\sigma(q)$)

Under C2, the correction $\sigma(q)$ must be derived from the representation structure of the Heisenberg group. The Weil representation has a metaplectic phase structure ($\text{SL}_2(\mathbb{Z}/q\mathbb{Z})$ acts via Weil’s representation) that may contribute a systematic correction to the growth dynamics.

O15-O4 (Quark and neutrino sectors)

Both scenarios modify the $\delta \rightarrow \beta^*$ map. The extension to the quark and neutrino sectors (O14-O2 [19]) requires re-evaluating the corrected relation at the block structures appropriate for those sectors.

9 Conclusion

The O3–O7 derivation chain that produces $\beta^* = 1/(\delta + 1/2)$ is logically correct in its domain: a scalar global capacity on proxy-level LPS dynamics. The failure is not an error but a scope limitation: the scalar-observable hypothesis H-O6-1 does not hold for the exact Weil-block capacity measured in O12–O14.

The non-equivalence theorem (Proposition 3.2) establishes that the block-averaged exponent $\hat{\delta}_{\text{exact}}$ does not coincide with the scalar exponent α entering the growth equation, and that the bias goes in the wrong direction (Remark 3.3).

The effective dynamical redundancy R_n^{eff} (Definition 4.1) is the correct observable entering the growth equation in the exact setting. Its exponent α_{dyn} is smaller than $\hat{\delta}_{\text{exact}}$, so the aggregation correction alone cannot resolve the tension.

The aggregation no-go (Corollary 4.5) is the sharpest negative result: no reweighting of the block-level data can bring the dynamical exponent into the target window. The resolution must come from the growth equation itself.

The re-derivation of the growth equation (Hypothesis 5.2) reveals that the passage from proxy to exact introduces two additional corrections: the shell growth dimension $(D - 1)$ and the block-normalisation $\log q / \log n^*$. These corrections reduce the effective exponent and bring β^* closer to the target, but under Scenario C1 the residual gap is still a factor of 3.

Scenario C2, which introduces a q -dependent structural correction $\sigma(q)$, can close the gap but requires $\sigma(q) \sim 5$, a large value whose derivation from the Weil representation structure is an open problem.

The minimal next step is the block-level redundancy extraction (O15-O1): computing R_n^{eff} from existing O12/O13 data to measure α_{dyn} directly and discriminate the two scenarios. This measurement is feasible with the current pipeline outputs and constitutes the critical path forward.

The O-series has now traced the chain from obstruction identification (O8–O9) through exact measurement (O10–O13), theoretical correspondence (O14), and derivation audit (O15). The status of Source S2 is transformed: it is no longer an unexplained numerical tension, nor a theoretical impasse, but a precisely formulated computational problem. The gap between $\hat{\delta}_{\text{exact}}$ and the target window is quantified, its origin is localised at the growth-equation level, and the discriminating measurement (O15-O1: extraction of α_{dyn} from existing pipeline data) is immediately feasible. O15 thus converts the open question “why does δ not match β^* ?” into the concrete programme “measure R_n^{eff} , test C1, and if C1 fails, derive $\sigma(q)$ from the Weil representation structure.”

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